NIOS Class 10 (211) Mathematics — 7-Day Fast-Track Study Plan

NIOS Secondary (Class 10) Mathematics (211) — Exam Prep Study Kit

NIOS Class 10 (211) Mathematics — 7-Day Fast-Track Study Plan

Designed for Working Students | Exam Date: 5th November

> **Target**: Pass comfortably with **50+ to 65+ marks** out of 85.
> **Daily Time Required**: 2 hours on weekdays (1 hr morning + 1 hr night), 3–4 hours on weekend.

The Working Student's Realistic Routine

- **Morning Sprint (45–60 mins)**: Review formulas, learn 1 high-yield template, solve 5 MCQs.
- **Commute / Breaks (15–20 mins)**: Glance through the `02_ALL_IN_ONE_FORMULA_CHEAT_SHEET.md` on your phone.
- **Evening Session (60–90 mins)**: Practice the target chapter's solved questions with pen and paper.

Day-by-Day Roadmap (Oct 30 – Nov 5)

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graph TD
    D1["Day 1 (Oct 30): Guaranteed 17 Marks<br>Heights & Distances + Trig Values + Probabilit
    D2 --> D3["Day 3 (Nov 1): Geometry & Graph (15 Marks)<br>Graphical Equations (5M) + Circl
    D3 --> D4["Day 4 (Nov 2): Statistics & Coordinate Geom (15 Marks)<br>Mean/Median Formula
    D4 --> D5["Day 5 (Nov 3): Mensuration & Commercial (12 Marks)<br>Cylinder/Cone/Sphere + I
    D5 --> D6["Day 6 (Nov 4): Objective Blitz & Mock Test<br>Practice Top 50 Objectives (42M)
    D6 --> D7["Day 7 (Nov 5): Exam Day Final Warm-Up<br>Formula Cheat Sheet Review & Confiden
```

Day 1 (Friday, Oct 30): The Easiest 17 Marks

- **Topics**:
- **Trigonometric Values Table** ($\sin, \cos, \tan$ for $0^\circ, 30^\circ, 45^\circ, 60^\circ, 90^\circ$).
- **Heights & Distances (5 Marks Sure-Shot!)**: Tower, light house, angle of elevation/depression.
- **Probability (Ch 26 - 5 Marks)**: Coins, 1 or 2 dice, cards, balls in a bag.
- **Goal**: Complete these 2 topics tonight. You will have locked in **17 marks** on Day 1!
- **Read**:
- `04_GUARANTEED_SURE_SHOT_LONG_QUESTIONS.md` (Template 1: Heights & Distances)
- `02_ALL_IN_ONE_FORMULA_CHEAT_SHEET.md` (Trigonometry & Probability section)

Day 2 (Saturday, Oct 31): Algebra Powerhouse (15 Marks)

- **Topics**:
- **Arithmetic Progression (AP - Ch 7)**:
  - n-th term: $a_n = a + (n-1)d$
  - Sum: $S_n = \frac{n}{2}[2a + (n-1)d]$
- **Quadratic Equations (Ch 6)**:
  - Discriminant $D = b^2 - 4ac$ (Nature of roots: $D > 0, D = 0, D < 0$)
  - Finding roots using formula $x = \frac{-b \pm \sqrt{D}}{2a}$
- **Special Products & Factorisation (Ch 4)**:
  - $(a+b)^2, (a-b)^2, a^2 - b^2, a^3 + b^3, a^3 - b^3$.
- **Goal**: Solve the 8 standard problems in `05_CHAPTER_BY_CHAPTER_SUPER_SUMMARY.md`.

Day 3 (Sunday, Nov 1 - Weekend Deep Work): The 5-Mark Graph & Geometry (15 Marks)

- **Topics**:
- **Linear Equations Graphical Solution (5 Marks Sure-Shot!)**:
  - Plotting $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$ on a graph sheet, finding intersection point (x, y).
- **Circle Tangents & Angles (Ch 16 & 17)**:
  - Tangent $\perp$ radius ($OP \perp PT$).
  - Tangents from external point are equal ($PA = PB$).
  - Angle in semicircle is 90°; Cyclic quadrilateral opposite angles sum to 180°.
- **Construction (Ch 18 - 3 to 5 Marks)**:
  - Tangents to a circle from an external point using compass and ruler.
- **Goal**: Practice drawing 2 graphs and 2 circle tangent constructions on paper!

Day 4 (Monday, Nov 2): Statistics & Coordinate Geometry (15 Marks)

- **Topics**:
- **Statistics (Ch 25)**:
  - Grouped data Median: $L + \left(\frac{\frac{N}{2} - cf}{f} \right) \times h$
  - Grouped data Mean: $\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$
  - Empirical relation: $\text{Mode} = 3\text{Median} - 2\text{Mean}$
- **Coordinate Geometry (Ch 19)**:
  - Distance formula: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
  - Mid-point formula: $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$
  - Section formula: $\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$
- **Goal**: Solve 2 median problems and 3 coordinate geometry problems.

Day 5 (Tuesday, Nov 3): Mensuration & Commercial Maths (12 Marks)

- **Topics**:
- **Surface Area & Volume (Ch 21)**:
  - Cylinder: $V = \pi r^2 h$, $\text{CSA} = 2\pi rh$, $\text{TSA} = 2\pi r(r + h)$
  - Cone: $V = \frac{1}{3}\pi r^2 h$, $l = \sqrt{r^2 + h^2}$, $\text{CSA} = \pi rl$
  - Sphere & Hemisphere: $V = \frac{4}{3}\pi r^3$, $\text{TSA} = 4\pi r^2$
- **Installment Buying (Ch 9)**:
  - Cash price, Down payment, Balance owed, Interest calculation.
- **Goal**: Master formula substitutions for direct 2-mark and 3-mark questions.

Day 6 (Wednesday, Nov 4): The Objective Blitz & Rehearsal (Crucial Day!)

- **Topics**:
- Go through `03_TOP_50_REPEATED_OBJECTIVES_MCQ_SOLVED.md` from start to finish.
- Attempt the 85-mark `06_PREDICTED_MOCK_TEST_PAPER_2026.md` under timed conditions (or self-check answers).
- Pack your exam kit: Admit card/hall ticket, ID proof, Geometry box (ruler, compass, sharp pencil, eraser), black/blue pens.

Day 7 (Thursday, Nov 5 - Exam Day Morning):

- **45-minute calm morning revision**:
- Read through `02_ALL_IN_ONE_FORMULA_CHEAT_SHEET.md`.
- Remember: You only need 28 to pass, but your target is 50–65+.
- In the exam room: First 15 minutes, tick all the objective questions you know directly, then solve the 5-mark Heights & Distance and Graphical Linear equation. Walk in with full confidence!

NIOS Class 10 (211) Mathematics — Syllabus Bifurcation & Exam Blueprint

NIOS Secondary (Class 10) Mathematics (211) — Exam Prep Study Kit

NIOS Class 10 (211) Mathematics — Syllabus Bifurcation & Exam Blueprint

> **CRITICAL TIME-SAVING RULE FOR NOVEMBER 5 EXAM:**
> NIOS has officially bifurcated the syllabus into **TMA (40%)** and **Public Examination (60%)**. **DO NOT waste your precious time studying TMA chapters.** TMA chapters are evaluated ONLY in assignments and **will NOT be asked in the Public Examination**.

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1. What to Study vs What to Completely Skip

Module	Total Chapters	❌ DO NOT STUDY (TMA Only - 40%)	✅ MUST STUDY (Public Exam - 60%)	Module Marks Weightage
Module 1: Algebra	7	• Ch 1: Number System • Ch 2: Exponents & Radicals • Ch 3: Algebraic Expressions	• **Ch 4: Special Product & Factorisation** • **Ch 5: Linear Equations** • **Ch 6: Quadratic Equations** • **Ch 7: Arithmetic Progression (AP)**	**20 Marks**
Module 2: Commercial Maths	2	• Ch 8: Percentage & Its Applications	• **Ch 9: Installment Buying**	**8 Marks**
Module 3: Geometry	10	• Ch 10: Lines & Angles • Ch 11: Congruence of Triangles • Ch 12: Concurrent Lines • Ch 13: Quadrilaterals • Ch 14: Similarity of Triangles • Ch 15: Circles	• **Ch 16: Angles in Circle & Cyclic Quads** • **Ch 17: Tangents, Secants & Properties** • **Ch 18: Constructions** • **Ch 19: Coordinate Geometry**	**25 Marks**
Module 4: Mensuration	2	• Ch 20: Perimeter & Area of Plane Figures	• **Ch 21: Surface Area and Volume**	**10 Marks**
Module 5: Trigonometry	2	• Ch 22: Introduction to Trigonometry	• **Ch 23: Trigonometric Ratios & Angles (Heights & Distances)**	**10 Marks**

Module	Total Chapters	❌ DO NOT STUDY (TMA Only - 40%)	✅ MUST STUDY (Public Exam - 60%)	Module Marks Weightage
Module 6: Statistics & Probability	3	• Ch 24: Data & Their Representation	**• Ch 25: Measure of Central Tendency (Mean, Median)** **• Ch 26: Probability**	**12 Marks**
TOTAL	**26 Lessons**	**13 Lessons SKIPPED (Save 50% Time!)**	**13 High-Yield Lessons ONLY**	**85 Marks**

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2. Question Paper Pattern (85 Marks, 2½ Hours)

The exam paper contains **44 questions** divided into two sections:

Section A: Objective Questions — 42 Marks (49.4% of the Paper!)

- **Q1 to Q17 (17 Questions × 1 Mark = 17 Marks)**: Multiple Choice Questions (MCQs). Direct formula values, basic concepts.
- **Q18 to Q27 (10 Questions × 2 Marks = 20 Marks)**: Objective Type (each question has two 1-mark subparts: Fill in the blanks, True/False, Match the column).
- **Q28 (1 Question × 5 Marks = 5 Marks)**: Case Study / Context question with five 1-mark subparts.

> **Key Strategy**: Scoring 35+ out of these 42 objective marks immediately guarantees passing with flying colours!

Section B: Subjective Questions — 43 Marks

- **Q29 to Q37 (9 Questions × 2 Marks = 18 Marks)**: Very Short Answer (VSA). (4 have internal choices).
- **Q38 to Q42 (5 Questions × 3 Marks = 15 Marks)**: Short Answer (SA). (2 have internal choices).
- **Q43 to Q44 (2 Questions × 5 Marks = 10 Marks)**: Long Answer (LA). (Both have internal choices).

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3. The "Sure-Shot" 30 Marks Checklist

If you master just these 6 topics, you will easily secure 30–35 marks in Section B alone:

1. **Heights & Distances (Trigonometry) [5 Marks]**:
2. Always Q44 (Light House / Tower / Building with angles $30^\circ, 45^\circ, 60^\circ$).
3. **Graphical Solution of Linear Equations OR Circle Tangents Construction [5 Marks]**:
4. Always Q43 (Plot two straight lines on graph sheet to find intersection, or construct tangents to circle from external point).
5. **Median / Mean of Grouped Data [3 Marks]**:
6. Always Q41 or Q42 (Using the median formula $L + \frac{\frac{N}{2} - cf}{f} \times h$).
7. **Coordinate Geometry [3 to 5 Marks]**:
8. Distance formula, Mid-point formula, and Section formula dividing line segment.
9. **Arithmetic Progression (AP) [3 to 4 Marks]**:
10. Finding the n-th term ($a_n = a + (n-1)d$) and sum of n terms (S_n).
11. **Probability [3 Marks]**:
12. Simple problems on single/double dice, coin toss, or deck of 52 cards.

NIOS Class 10 (211) Mathematics — Master Formula Cheat Sheet

NIOS Secondary (Class 10) Mathematics (211) — Exam Prep Study Kit

NIOS Class 10 (211) Mathematics — Master Formula Cheat Sheet

1-Stop Formula Reference for Public Examination

> **How to use this**: Keep this file open on your phone or print it out. Review these formulas every single day until 5th November. Over 60% of exam marks come from direct application of these formulas!

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1. Trigonometry & Heights and Distances (Module 5 — 10 Marks)

Trigonometric Values Table (MEMORIZE THIS ENTIRE ROW!)

Angle (θ)	0°	30°	45°	60°	90°
$\sin\theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan\theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	Not defined
$\operatorname{cosec}\theta$	Not defined	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1
$\sec\theta$	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	Not defined
$\cot\theta$	Not defined	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0

> **Pro Tip for Heights & Distances (5 Marks Q44)**: In almost every problem, you will use $\tan\theta = \frac{\text{Height (Opposite)}}{\text{Distance (Adjacent)}}$: - When $\theta = 45^\circ \implies \tan 45^\circ = 1 \implies \text{Height} = \text{Distance}$! - When $\theta = 30^\circ \implies \tan 30^\circ = \frac{1}{\sqrt{3}}$ - When $\theta = 60^\circ \implies \tan 60^\circ = \sqrt{3}$ - Remember: $\sqrt{3} \approx 1.732$.

Core Trigonometric Identities

- $\sin^2\theta + \cos^2\theta = 1 \implies \sin^2\theta = 1 - \cos^2\theta \implies \cos^2\theta = 1 - \sin^2\theta$
- $1 + \tan^2\theta = \sec^2\theta \implies \sec^2\theta - \tan^2\theta = 1$
- $1 + \cot^2\theta = \text{cosec}^2\theta \implies \text{cosec}^2\theta - \cot^2\theta = 1$
- $\tan\theta = \frac{\sin\theta}{\cos\theta}, \quad \cot\theta = \frac{\cos\theta}{\sin\theta}, \quad \sec\theta = \frac{1}{\cos\theta}, \quad \text{cosec}\theta = \frac{1}{\sin\theta}$

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2. Algebra (Module 1 — 20 Marks)

Special Products & Factorisation (Ch 4)

- $(a+b)^2 = a^2 + 2ab + b^2$
- $(a-b)^2 = a^2 - 2ab + b^2$
- $(a+b)^2 + (a-b)^2 = 2(a^2 + b^2)$
- $(a+b)^2 - (a-b)^2 = 4ab$
- $a^2 - b^2 = (a+b)(a-b)$
- $a^3 + b^3 = (a+b)(a^2 - ab + b^2)$
- $a^3 - b^3 = (a-b)(a^2 + ab + b^2)$
- $(a+b+c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$

Linear Equations in Two Variables (Ch 5)

For system $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$: - $\frac{a_1}{a_2} \neq \frac{b_1}{b_2} \implies$ **Unique solution** (Lines intersect at 1 point, Consistent). - $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} \implies$ **Infinitely many solutions** (Coincident lines, Dependent Consistent). - $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \implies$ **No solution** (Parallel lines, Inconsistent).

Quadratic Equations (Ch 6)

Standard form: $ax^2 + bx + c = 0$ $(a \neq 0)$ - **Discriminant**: $D = b^2 - 4ac$ - **Nature of Roots**: - $D > 0 \implies$ Two real and distinct roots - $D = 0 \implies$ Two real and equal roots ($x = -\frac{b}{2a}$) - $D <$

$0 \implies$ No real roots - **Quadratic Formula**:

$$x = \frac{-b \pm \sqrt{D}}{2a} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- **Sum & Product of Roots**: - Sum of roots $\alpha + \beta = -\frac{b}{a}$ - Product of roots $\alpha \cdot \beta = \frac{c}{a}$

Arithmetic Progression (AP - Ch 7)

- Common difference: $d = a_n - a_{n-1}$
- n**-th term**:

$$a_n = a + (n-1)d$$

- **Sum of first** n **terms**:

$$S_n = \frac{n}{2}[2a + (n-1)d] \quad \text{or} \quad S_n = \frac{n}{2}[a + l]$$

(where $l = a_n$ *is the last term)*

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3. Coordinate Geometry (Module 3 — Ch 19 — 6 to 8 Marks)

- **Distance between two points** $P(x_1, y_1)$ **and** $Q(x_2, y_2)$:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

- **Distance of point** (x, y) **from origin** $(0, 0)$:

$$d = \sqrt{x^2 + y^2}$$

- **Mid-point of line segment joining** (x_1, y_1) **and** (x_2, y_2):

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

- **Section Formula (Internal division in ratio** $m : n$**)**:

$$P(x, y) = \left(\frac{mx_2 + nx_1}{m + n}, \frac{my_2 + ny_1}{m + n} \right)$$

(Tip: If point is on y-axis, $x = 0$*. If point is on x-axis,* $y = 0$*)*

- **Centroid of a Triangle with vertices** $(x_1, y_1), (x_2, y_2), (x_3, y_3)$:

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

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4. Circle Geometry & Theorems (Module 3 — Ch 16 & 17 — 10 Marks)

1. **Tangent $\perp$ Radius**: A tangent at any point of a circle is perpendicular to the radius through the point of contact ($\angle OPT = 90^\circ$).
2. **Lengths of Tangents**: The lengths of two tangents drawn from an external point to a circle are equal ($PA = PB$).
3. **Angle Subtended at Centre**: The angle subtended by an arc at the centre is twice the angle subtended by it at any point on the remaining part of the circle ($\angle AOC = 2\angle ABC$).
4. **Angle in Semicircle**: Angle in a semicircle is a right angle (90°).
5. **Angles in Same Segment**: Angles in the same segment of a circle are equal.
6. **Cyclic Quadrilateral**: The sum of either pair of opposite angles of a cyclic quadrilateral is 180° ($\angle A + \angle C = 180^\circ,\ \angle B + \angle D = 180^\circ$).

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5. Mensuration (Module 4 — Ch 21 — 10 Marks)

Solid Shape	Curved Surface Area (CSA)	Total Surface Area (TSA)	Volume (V)
Cuboid (l, b, h)	$2h(l + b)$ *(Area of 4 walls)*	$2(lb + bh + hl)$	$l \times b \times h$
Cube (side a)	$4a^2$	$6a^2$	a^3
Cylinder (radius r, height h)	$2\pi rh$	$2\pi r(r + h)$	$\pi r^2 h$
Cone (radius r, height h, slant l)	πrl *(where $l = \sqrt{r^2 + h^2}$)*	$\pi r(r + l)$	$\frac{1}{3}\pi r^2 h$
Sphere (radius r)	$4\pi r^2$	$4\pi r^2$	$\frac{4}{3}\pi r^3$
Hemisphere (radius r)	$2\pi r^2$	$3\pi r^2$	$\frac{2}{3}\pi r^3$

- **Circle Sector & Ring**:

- Area of circle $= \pi r^2$, Circumference $= 2\pi r$
- Area of Sector with central angle $\theta = \frac{\theta}{360^\circ} \times \pi r^2$
- Length of Arc $= \frac{\theta}{360^\circ} \times 2\pi r$
- Area of Circular Ring $= \pi(R^2 - r^2) = \pi(R + r)(R - r)$

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6. Statistics & Probability (Module 6 — Ch 25 & 26 — 12 Marks)

Statistics (Ch 25)

- **Mean (Direct Method)**:

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$$

- **Median of Grouped Data (3 Marks Sure-Shot!)**:

$$\text{Median} = L + \left(\frac{\frac{N}{2} - cf}{f} \right) \times h$$

- L = Lower limit of median class
- $N = \sum f_i$ = Total frequency
- cf = Cumulative frequency of the class *preceding* the median class
- f = Frequency of the median class
- h = Class width (upper limit − lower limit)
- **Empirical Formula (Very common 1-mark question!)**:

$$\text{Mode} = \mathbf{3}\,\text{Median} - \mathbf{2}\,\text{Mean}$$

Probability (Ch 26)

- $$P(E) = \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}}$$

- Range of probability: $0 \leq P(E) \leq 1$
- Sum of complementary probabilities: $P(E) + P(\bar{E}) = 1 \implies P(\text{not } E) = 1 - P(E)$
- Standard coin toss outcomes:
- 1 coin: $\{H, T\}$ (Total = 2)
- 2 coins: $\{HH, HT, TH, TT\}$ (Total = 4)
- Standard dice outcomes:

- 1 die: $\{1, 2, 3, 4, 5, 6\}$ (Total = 6)
- 2 dice: Total outcomes = $6 \times 6 = 36$
- Standard deck of cards: Total = 52 cards (26 Red: 13 Hearts + 13 Diamonds; 26 Black: 13 Spades + 13 Clubs; 12 Face Cards: 4 Kings + 4 Queens + 4 Jacks; 4 Aces).

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7. Commercial Mathematics — Installment Buying (Ch 9 — 8 Marks)

- **Cash Price**: Price of the article if paid in full immediately.
- **Cash Down Payment (CDP)**: Amount paid in cash at the time of purchase.
- **Unpaid Balance (Principal)** $= \text{Cash Price} - \text{Cash Down Payment}$
- **Total Amount Paid under Installment Scheme** $= \text{Down Payment} + (n \times \text{Monthly Installment Amount})$
- **Total Interest Paid** $= \text{Total Amount Paid} - \text{Cash Price}$

NIOS Class 10 (211) Mathematics — Top 50 Most Repeated Objective Questions (MCQs, Blanks & True/False)

NIOS Secondary (Class 10) Mathematics (211) — Exam Prep Study Kit

NIOS Class 10 (211) Mathematics — Top 50 Most Repeated Objective Questions (MCQs, Blanks & True/False)

Target: Master the 42 Objective Marks in Section A

> **Why this file is your biggest asset:**
> In NIOS Mathematics 211, **Section A carries 42 marks out of 85** (nearly 50% of the entire paper!).
> If you score 35+ on these quick, 1-step objective questions, you have already guaranteed a passing score before even opening Section B!

Part 1: Multiple Choice Questions (1 Mark Each)

1. Quadratic Equations & Algebra

Q1. The discriminant of the quadratic equation $x^2 - 4x + 3 = 0$ is:
(A) 4
(B) -4
(C) 2
(D) 0

> **Answer: (A) 4**
> *Explanation*: $D = b^2 - 4ac = (-4)^2 - 4(1)(3) = 16 - 12 = 4$. Since $D > 0$, roots are real and distinct.

Q2. If the quadratic equation $kx^2 - 4x + 1 = 0$ has two equal real roots, then the value of k is:

(A) 2

(B) 4

(C) -4

(D) 16

> **Answer: (B) 4**
> *Explanation*: For equal roots, $D = b^2 - 4ac = 0 \implies (-4)^2 - 4(k)(1) = 0 \implies 16 = 4k \implies k = 4$.

Q3. The roots of the quadratic equation $x^2 - 25 = 0$ are:

(A) 5

(B) -5

(C) ± 5

(D) 25

> **Answer: (C) ± 5**
> *Explanation*: $x^2 = 25 \implies x = \pm\sqrt{25} = \pm 5$.

Q4. The 10th term of the AP $2, 7, 12, 17, \ldots$ is:

(A) 45

(B) 47

(C) 50

(D) 52

> **Answer: (B) 47**
> *Explanation*: $a = 2, d = 7 - 2 = 5$. Formula: $a_n = a + (n-1)d \implies a_{10} = 2 + (10-1) \times 5 = 2 + 45 = 47$.

Q5. In an AP, if $a = 5, d = 3$, and $a_n = 50$, then n is:

(A) 14

(B) 15

(C) 16

(D) 17

Answer: (C) 16
Explanation: $50 = 5 + (n-1) \times 3 \implies 45 = 3(n-1) \implies n - 1 = 15 \implies n = 16.$

Q6. The sum of first 10 terms of the AP $1, 3, 5, 7, \ldots$ is:
(A) 100
(B) 90
(C) 110
(D) 50

Answer: (A) 100
Explanation: The sum of the first n odd natural numbers is $n^2 \implies 10^2 = 100.$

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2. Geometry & Coordinate Geometry

Q7. The distance between the points (a, b) and $(-a, -b)$ is:
(A) $\sqrt{a^2 + b^2}$
(B) $2\sqrt{a^2 + b^2}$
(C) $a + b$
(D) $2(a + b)$

Answer: (B) $2\sqrt{a^2 + b^2}$
Explanation: $d = \sqrt{(-a - a)^2 + (-b - b)^2} = \sqrt{(-2a)^2 + (-2b)^2} = \sqrt{4a^2 + 4b^2} = 2\sqrt{a^2 + b^2}.$

Q8. The mid-point of the line segment joining the points $(-1, 3)$ and $(3, 5)$ is:
(A) $(1, 4)$
(B) $(2, 4)$
(C) $(1, 8)$
(D) $(2, 8)$

Answer: (A) (1, 4)
Explanation: $M = \left(\frac{-1+3}{2}, \frac{3+5}{2}\right) = \left(\frac{2}{2}, \frac{8}{2}\right) = (1, 4).$

Q9. The distance of the point $P(3, 4)$ from the origin $(0, 0)$ is:
(A) 3

(B) 4
(C) 5
(D) 7

> **Answer: (C) 5**
> *Explanation*: $d = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5.$

Q10. The length of tangent from an external point A to a circle of radius $7\ \text{cm}$ is $24\ \text{cm}$. The distance of point A from the centre is:
(A) $25\ \text{cm}$
(B) $31\ \text{cm}$
(C) $17\ \text{cm}$
(D) $26\ \text{cm}$

> **Answer: (A) 25 cm**
> *Explanation*: By Pythagoras theorem, $d = \sqrt{r^2 + \text{tangent}^2} = \sqrt{7^2 + 24^2} = \sqrt{49 + 576} = \sqrt{625} = 25\ \text{cm}.$

Q11. The angle subtended by a diameter at any point on the semicircle is:
(A) 45°
(B) 60°
(C) 90°
(D) 180°

> **Answer: (C) 90°**
> *Explanation*: Angle in a semicircle is always a right angle (90°).

Q12. In a cyclic quadrilateral $ABCD$, if $\angle A = 70^\circ$, then the measure of $\angle C$ is:
(A) 70°
(B) 110°
(C) 90°
(D) 120°

> **Answer: (B) 110°**
> *Explanation*: Opposite angles of a cyclic quadrilateral sum to $180^\circ \implies \angle C = 180^\circ - 70^\circ = 110^\circ.$

3. Trigonometry

Q13. The value of $\sec 60^\circ$ is:
(A) $\frac{1}{2}$
(B) $\frac{\sqrt{3}}{2}$
(C) $\sqrt{2}$
(D) 2

> **Answer: (D) 2**
> *Explanation*: $\cos 60^\circ = \frac{1}{2} \implies \sec 60^\circ = \frac{1}{\cos 60^\circ} = 2.$

Q14. The value of $\sin^2 30^\circ + \cos^2 30^\circ$ is:
(A) 0
(B) 1
(C) $\frac{1}{2}$
(D) 2

> **Answer: (B) 1**
> *Explanation*: Identity $\sin^2 \theta + \cos^2 \theta = 1$ holds for all angles.

Q15. If $\tan \theta = 1$, then the value of θ is:
(A) 30°
(B) 45°
(C) 60°
(D) 90°

> **Answer: (B) 45°**
> *Explanation*: $\tan 45^\circ = 1.$

Q16. The value of $\tan 60^\circ \cdot \cot 60^\circ$ is:
(A) $\sqrt{3}$
(B) $\frac{1}{\sqrt{3}}$
(C) 1
(D) 0

Answer: (C) 1
Explanation: $\cot\theta = \frac{1}{\tan\theta} \implies \tan\theta \cdot \cot\theta = 1.$

---

4. Mensuration

Q17. Area of a square whose perimeter is $44\ \text{cm}$ is:
(A) $121\ \text{cm}^2$
(B) $144\ \text{cm}^2$
(C) $44\ \text{cm}^2$
(D) $169\ \text{cm}^2$

Answer: (A) 121 cm²
Explanation: $\text{Perimeter} = 4a = 44 \implies a = 11\ \text{cm} \implies \text{Area} = a^2 = 11^2 = 121\ \text{cm}^2.$

Q18. The volume of a cylinder of the same base radius and height as that of a cone is:
(A) Equal to volume of cone
(B) 2 times the volume of cone
(C) 3 times the volume of cone
(D) $\frac{1}{3}$ the volume of cone

Answer: (C) 3 times the volume of cone
Explanation: $V_{\text{cylinder}} = \pi r^2 h = 3 \times \left(\frac{1}{3}\pi r^2 h\right) = 3 \times V_{\text{cone}}.$

Q19. Length, breadth, and height of a cuboid are $3\ \text{cm}, 4\ \text{cm}$, and $5\ \text{cm}$ respectively. Its volume is:
(A) $12\ \text{cm}^3$
(B) $60\ \text{cm}^3$
(C) $20\ \text{cm}^3$
(D) $48\ \text{cm}^3$

Answer: (B) 60 cm³
Explanation: $V = l \times b \times h = 3 \times 4 \times 5 = 60\ \text{cm}^3.$

Q20. If the radius of a sphere is doubled, its surface area becomes:
(A) 2 times
(B) 3 times

(C) 4 times
(D) 8 times

> **Answer: (C) 4 times**
> *Explanation*: Surface Area $= 4\pi r^2$. When $r \to 2r$, $\text{New Area} = 4\pi(2r)^2 = 4 \times (4\pi r^2)$.

---

5. Statistics & Probability

Q21. The middle-most observation of a data set arranged in ascending or descending order is called:
(A) Mean
(B) Median
(C) Mode
(D) Range

> **Answer: (B) Median**

Q22. If Mean $= 10$ and Median $= 12$, then Mode using the empirical formula is:
(A) 14
(B) 16
(C) 18
(D) 20

> **Answer: (B) 16**
> *Explanation*: $\text{Mode} = 3\,\text{Median} - 2\,\text{Mean} = 3(12) - 2(10) = 36 - 20 = 16$.

Q23. If $P(E) = 0.05$, then the probability of 'not E' is:
(A) 0.95
(B) 0.05
(C) 1.05
(D) 0

> **Answer: (A) 0.95**
> *Explanation*: $P(\bar{E}) = 1 - P(E) = 1 - 0.05 = 0.95$.

Q24. If $P(E) + P(\bar{E}) = y$, then the value of $y^3 - 4$ is:
(A) 4

(B) 3
(C) -3
(D) 0

> **Answer: (C) -3**
> *Explanation*: Since $P(E) + P(\bar{E}) = 1 \implies y = 1 \implies y^3 - 4 = 1^3 - 4 = -3$. *(Very popular NIOS question!)*

Q25. Two coins are tossed simultaneously. The probability of getting at least one head is:
(A) $\frac{1}{4}$
(B) $\frac{1}{2}$
(C) $\frac{3}{4}$
(D) 1

> **Answer: (C) 3/4**
> *Explanation*: Sample space $= \{HH, HT, TH, TT\}$ (4 outcomes). Favourable $= \{HH, HT, TH\}$ (3 outcomes) $\implies \frac{3}{4}$.

Q26. Two dice are thrown together. The probability that they show different numbers is:
(A) $\frac{1}{6}$
(B) $\frac{5}{6}$
(C) $\frac{1}{36}$
(D) $\frac{35}{36}$

> **Answer: (B) 5/6**
> *Explanation*: Total outcomes $= 36$. Same numbers (doublets) $= \{(1,1), (2,2), (3,3), (4,4), (5,5), (6,6)\} = 6 \implies P(\text{same}) = \frac{6}{36} = \frac{1}{6}$. Therefore, $P(\text{different}) = 1 - \frac{1}{6} = \frac{5}{6}$.

Q27. A card is drawn from a well-shuffled deck of 52 cards. The probability that it is an Ace is:
(A) $\frac{1}{52}$
(B) $\frac{1}{13}$
(C) $\frac{4}{13}$
(D) $\frac{1}{4}$

Answer: (B) 1/13
Explanation: Total Aces $= 4 \implies P = \frac{4}{52} = \frac{1}{13}$.

Part 2: Fill in the Blanks & Match Questions (2 Marks Each / 1 Mark Per Part)

Q28.
(i) Two factors $(x + 3)$ and _____ are obtained on factorising $x^2 + 8x + 15$.

Answer: $(x + 5)$
Reason: $3 \times 5 = 15$ and $3 + 5 = 8$.

(ii) The value of the polynomial $2x + 3x^2 - 4$ at $x = 0$ is _____.

Answer: -4
Reason: $2(0) + 3(0)^2 - 4 = -4$.

Q29.
(i) Angles in the same segment of a circle are _____.

Answer: **equal**

(ii) If the sum of a pair of opposite angles of a quadrilateral is 180°, the quadrilateral is _____.

Answer: **cyclic**

Q30.
(i) A tangent to a circle intersects it in exactly _____ point(s).

Answer: **one**

(ii) A line intersecting a circle in two points is called a _____.

Answer: **secant**

Q31.

(i) The maximum number of parallel tangents that a circle can have is _____.

> **Answer: two**

(ii) The common point of a tangent to a circle and the circle is called _____.

> **Answer: point of contact**

Q32.

(i) $\sin^2\theta + \cos^2\theta =$ _____.

> **Answer:** 1

(ii) $1 + \tan^2\theta =$ _____.

> **Answer:** $\sec^2\theta$

Q33.

(i) The coordinates of the origin are _____.

> **Answer:** $(0, 0)$

(ii) The y-coordinate of any point on the x-axis is _____.

> **Answer:** 0

Q34.

(i) The probability of a sure (certain) event is _____.

> **Answer:** 1

(ii) The probability of an impossible event is _____.

> **Answer:** 0

Q35.

(i) If cash price of a cycle is ₹2,500 and down payment is ₹500, the unpaid balance is _____.

> **Answer**: **₹2,000** $(2500 - 500)$

(ii) Amount of each installment in 4 equal monthly installments for ₹2,000 (without interest) is _____.

> **Answer**: **₹500** $(2000/4)$

Part 3: True / False Statements (2 Marks Each / 1 Mark Per Part)

Q36.

(i) "Using a ruler and compass it is possible to construct an angle of 25°."

> **Answer**: **FALSE**
> *Reason*: Only multiples of 15° ($15^\circ, 30^\circ, 45^\circ, 60^\circ, 75^\circ, 90^\circ \ldots$) or their bisections ($7.5^\circ, 22.5^\circ$) can be constructed. 25° is not constructible.

(ii) "The tangent at any point of a circle is perpendicular to the radius through the point of contact."

> **Answer**: **TRUE**
> *Reason*: This is the fundamental Circle Tangent Theorem.

Q37.

(i) "Two distinct lines can have more than one point in common."

> **Answer**: **FALSE**
> *Reason*: Two distinct straight lines can intersect at at most 1 point.

(ii) "The ratio of the circumference to the diameter of any circle is always equal to π."

> **Answer**: **TRUE**
> *Reason*: $C = \pi d \implies \frac{C}{d} = \pi$.

Q38.

(i) "The probability of an event can be -0.5."

> **Answer**: **FALSE**
> *Reason*: Probability must always satisfy $0 \leq P(E) \leq 1$. It can never be negative.

(ii) "In an AP, the difference between any two consecutive terms is always constant."

> **Answer**: **TRUE**
> *Reason*: That is the definition of common difference d.

Q39.

(i) "The discriminant of the equation $x^2 + 4x + 5 = 0$ is negative."

> **Answer**: **TRUE**
> *Reason*: $D = 4^2 - 4(1)(5) = 16 - 20 = -4 < 0$.

(ii) "The value of $\sin 90^\circ$ is 0."

> **Answer**: **FALSE**
> *Reason*: $\sin 90^\circ = 1$ (while $\cos 90^\circ = 0$).

Q40.

(i) "The curved surface area of a cone is given by $\pi r l$."

> **Answer**: **TRUE**

(ii) "The volume of a sphere of radius r is $\frac{2}{3}\pi r^3$."

> **Answer**: **FALSE**
> *Reason*: Volume of a sphere is $\frac{4}{3}\pi r^3$ (Hemisphere is $\frac{2}{3}\pi r^3$).

NIOS Class 10 (211) Mathematics — Guaranteed Sure-Shot Questions & Solved Templates

NIOS Secondary (Class 10) Mathematics (211) — Exam Prep Study Kit

NIOS Class 10 (211) Mathematics — Guaranteed Sure-Shot Questions & Solved Templates

> **These 6 question templates appear in almost every single NIOS exam paper.**
> Mastering these 6 templates alone locks in **25 to 30 marks** in Section B! Practice writing each of these out by hand.

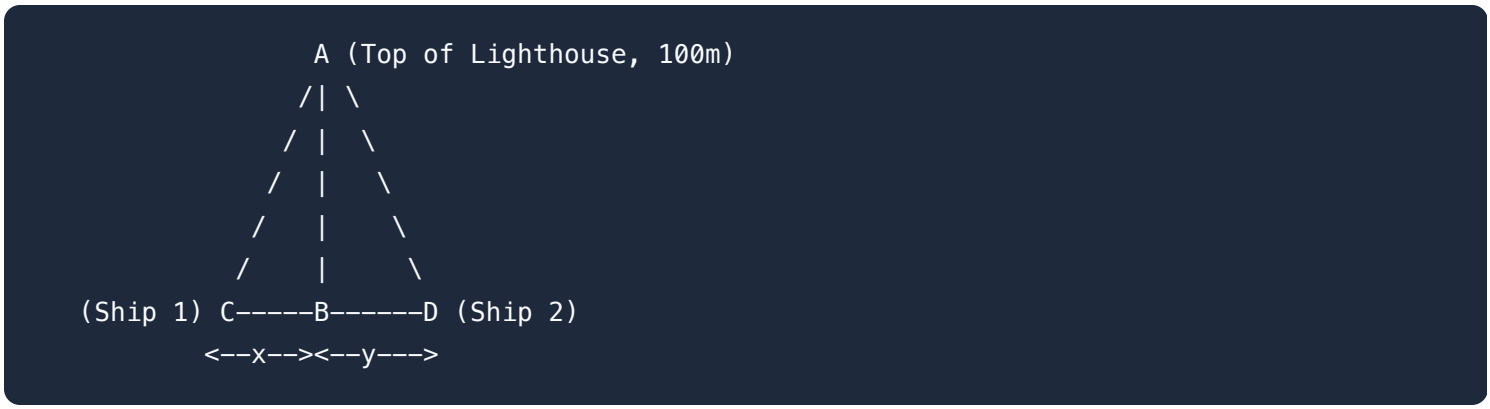
Template 1: Heights and Distances (5 Marks — Q44)

Problem (Real Exam Question from 2024 & 2025 Papers):

> As observed from the top of a $100\ \text{m}$ high lighthouse from the sea level, the angles of depression of two ships on the opposite sides of the lighthouse are 60° and 45° respectively. Find the distance between the two ships. (Take $\sqrt{3} \approx 1.732$).

Step-by-Step Solution:

1. **Draw the Diagram**:
2. Let AB be the lighthouse of height $h = 100\ \text{m}$.
3. Let C and D be the two ships on opposite sides on the sea level along a straight line.
4. The foot of the lighthouse is B.
5. Angle of elevation from ship C to top A is 45° (alternate interior angle to angle of depression).
6. Angle of elevation from ship D to top A is 60°.



1. **In right ΔABC**:

$$\tan 45^\circ = \frac{AB}{BC}$$

Since $\tan 45^\circ = 1$ and $AB = 100$:

$$1 = \frac{100}{BC} \implies BC = 100 \text{ m}$$

2. **In right ΔABD**:

$$\tan 60^\circ = \frac{AB}{BD}$$

Since $\tan 60^\circ = \sqrt{3}$ and $AB = 100$:

$$\sqrt{3} = \frac{100}{BD} \implies BD = \frac{100}{\sqrt{3}}$$

Rationalize the denominator:

$$BD = \frac{100\sqrt{3}}{3} = \frac{100 \times 1.732}{3} = \frac{173.2}{3} \approx 57.73 \text{ m}$$

3. **Total distance between the ships**:

$$\text{Distance } CD = BC + BD = 100 + 57.73 = \mathbf{157.73} \text{ m}$$

(Note: If ships were on the SAME side of the lighthouse, distance would be $BC - BD = 100 - 57.73 = 42.27$ m)

Template 2: Graphical Solution of Linear Equations (5 Marks — Q43)

Problem (Real Exam Question):

> Solve the following system of linear equations graphically:
>
> $$2x + y = 6$$
>
> $$2x - y = -2 \quad (\text{or } 2x - y + 2 = 0)$$

Step-by-Step Solution:

1. **Table of Values for Equation 1:** $2x + y = 6 \implies y = 6 - 2x$
2. If $x = 0 \implies y = 6 - 2(0) = 6 \implies (0, 6)$
3. If $x = 2 \implies y = 6 - 2(2) = 2 \implies (2, 2)$
4. If $x = 3 \implies y = 6 - 2(3) = 0 \implies (3, 0)$

x	0	2	3
y	6	2	0

1. **Table of Values for Equation 2:** $2x - y = -2 \implies y = 2x + 2$
2. If $x = 0 \implies y = 2(0) + 2 = 2 \implies (0, 2)$
3. If $x = 1 \implies y = 2(1) + 2 = 4 \implies (1, 4)$
4. If $x = -1 \implies y = 2(-1) + 2 = 0 \implies (-1, 0)$

x	0	1	-1
y	2	4	0

1. **Plotting on Graph Sheet**:
2. Plot points $(0, 6), (2, 2), (3, 0)$ and draw a straight line through them, labeling it $2x + y = 6$.
3. Plot points $(0, 2), (1, 4), (-1, 0)$ and draw a straight line through them, labeling it $2x - y = -2$.
4. **Point of Intersection**:

5. Both lines intersect at the point $P(1, 4)$.
6. **Conclusion**: The solution to the system is $\mathbf{x = 1,\ y = 4}$. *(Quick check:* $2(1) + 4 = 6$ ✓, $2(1) - 4 = -2$ ✓*)*

Template 3: Construction of Tangents to a Circle (3 to 5 Marks — Q39/Q43)

Problem:

> Draw a circle of radius $3\ \text{cm}$. From a point P at a distance of $7\ \text{cm}$ from the centre O, construct a pair of tangents to the circle. Measure the length of the tangents.

Step-by-Step Construction Steps (Write these exact steps in your answer sheet):

1. **Step 1**: Mark a point O as the centre and draw a circle of radius $3\ \text{cm}$ using a compass.
2. **Step 2**: Take a point P such that $OP = 7\ \text{cm}$. Join OP.
3. **Step 3**: Draw the perpendicular bisector of the line segment OP. Let M be the mid-point of OP ($OM = MP = 3.5\ \text{cm}$).
4. **Step 4**: With M as centre and radius MO (or MP), draw a circle that cuts the given circle at two points, say T_1 and T_2.
5. **Step 5**: Join PT_1 and PT_2.
6. PT_1 and PT_2 are the required pair of tangents from point P to the circle.

Mathematical Verification:

In right ΔOT_1P ($\angle OT_1P = 90^\circ$ because tangent $\perp$ radius):

$$PT_1 = \sqrt{OP^2 - OT_1^2} = \sqrt{7^2 - 3^2} = \sqrt{49 - 9} = \sqrt{40} \approx \mathbf{6.32}\ \text{cm}.$$

Template 4: Median of Grouped Frequency Distribution (3 Marks — Q42)

Problem (Real Exam Question):

> The distribution below gives the marks/weights of 30 students. Find the median:

Class Interval	$40-45$	$45-50$	$50-55$	$55-60$	$60-65$	$65-70$	$70-75$
Frequency (f)	2	3	8	6	6	3	2

Step-by-Step Solution:

1. **Prepare the Cumulative Frequency Table**:

Class Interval	Frequency (f)	Cumulative Frequency (cf)
$40-45$	2	2
$45-50$	3	$2+3=5$
$50-55$	8	$5+8=13$
$55-60$ **(Median Class)**	6 (f)	$13+6=19$
$60-65$	6	$19+6=25$
$65-70$	3	$25+3=28$
$70-75$	2	$28+2=30$
Total	$N=30$	

1. **Locate the Median Class**:

$$\frac{N}{2}=\frac{30}{2}=15$$

The cumulative frequency just greater than 15 is 19, which corresponds to the class interval $55 - 60$. Therefore, **Median Class** $= 55 - 60$.

2. **Identify the Formula Values**:

3. Lower limit of median class, $L = 55$
4. Class size, $h = 60 - 55 = 5$
5. Frequency of median class, $f = 6$

6. Cumulative frequency of preceding class, $cf = 13$

7. **Apply Median Formula**:

$$\text{Median} = L + \left(\frac{\frac{N}{2} - cf}{f} \right) \times h$$

$$\text{Median} = 55 + \left(\frac{15 - 13}{6} \right) \times 5 = 55 + \left(\frac{2}{6} \right) \times 5 = 55 + \frac{5}{3} = 55 + 1.67 = \mathbf{56.67}$$

Template 5: Coordinate Geometry Ratio & Section Formula (3 Marks — Q40)

Problem (Real Exam Question from 2025 Paper):

> Find the ratio in which the y-axis divides the line segment joining the points $A(4, -5)$ and $B(-1, 3)$. Also find the coordinates of the point of intersection.

Step-by-Step Solution:

1. **Set up the variables**:
2. Let the y-axis divide the segment AB in the ratio $k : 1$ at point P.

3. Any point on the y-axis has its x-coordinate equal to 0. So, let $P = (0, y)$.

4. **Apply Section Formula for the x-coordinate**:

$$x = \frac{k(x_2) + 1(x_1)}{k + 1}$$

Here $x_1 = 4,\ x_2 = -1,\ x = 0$:

$$0 = \frac{k(-1) + 1(4)}{k+1}$$

$$0 = -k + 4 \implies k = 4$$

Therefore, the required ratio is $k : 1 = \mathbf{4 : 1}$.

5. **Find the y-coordinate of P**:

$$y = \frac{k(y_2) + 1(y_1)}{k+1} = \frac{4(3) + 1(-5)}{4+1} = \frac{12-5}{5} = \frac{7}{5} = 1.4$$

6. **Conclusion**:

7. The ratio is $4 : 1$.
8. The coordinates of the point of intersection are $\left(\mathbf{0}, \frac{\mathbf{7}}{\mathbf{5}}\right)$.

Template 6: Circle Tangents Calculation (2 Marks — Q33)

Problem:

> From an external point P, two tangents PM and PN are drawn to a circle of radius $7\ \text{cm}$. If the distance of P from the centre O is $25\ \text{cm}$, find the length of each tangent.

Step-by-Step Solution:

1. Since the tangent to a circle is perpendicular to the radius through the point of contact, ΔOMP is a right-angled triangle with $\angle OMP = 90^\circ$.
2. By Pythagoras Theorem in right ΔOMP:

$$OP^2 = OM^2 + PM^2$$

$$25^2 = 7^2 + PM^2$$

$$625 = 49 + PM^2 \implies PM^2 = 625 - 49 = 576$$

$$PM = \sqrt{576} = \mathbf{24}\ \text{cm}$$

3. Since tangents from an external point are equal in length:

$$PN = PM = \mathbf{24}\ \text{cm}.$$

NIOS Class 10 (211) Mathematics — Chapter-by-Chapter Fast Revision

NIOS Secondary (Class 10) Mathematics (211) — Exam Prep Study Kit

NIOS Class 10 (211) Mathematics — Chapter-by-Chapter Fast Revision

> **Only the 13 Public Examination Chapters are included here.**
> Everything else in your textbook belongs to TMA and is excluded from the exam. Review these chapters one by one according to the 7-day study plan.

MODULE 1: ALGEBRA (20 Marks Total)

Chapter 4: Special Products and Factorisation

- **Key Concepts**:
- Identities: $(a \pm b)^2 = a^2 \pm 2ab + b^2$
- $(a + b)(a - b) = a^2 - b^2$
- $(a \pm b)^3 = a^3 \pm 3a^2b + 3ab^2 \pm b^3 = a^3 \pm b^3 \pm 3ab(a \pm b)$
- $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$
- $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$
- **Standard Solved Questions**:
- *Simplify* $(3x + 4y)^2 + (3x - 4y)^2$:
  - Use identity: $(a + b)^2 + (a - b)^2 = 2(a^2 + b^2)$.
  - Here $a = 3x, b = 4y \implies 2((3x)^2 + (4y)^2) = 2(9x^2 + 16y^2) = \mathbf{18x^2 + 32y^2}$.
- *Factorise* $a^6 - 64b^6$:
  - Write as difference of squares: $(a^3)^2 - (8b^3)^2 = (a^3 - 8b^3)(a^3 + 8b^3)$.
  - Now apply $x^3 - y^3$ and $x^3 + y^3$:
  - $a^3 - (2b)^3 = (a - 2b)(a^2 + 2ab + 4b^2)$
  - $a^3 + (2b)^3 = (a + 2b)(a^2 - 2ab + 4b^2)$

- **Final Answer**: $\mathbf{(a-2b)(a+2b)(a^2+2ab+4b^2)(a^2-2ab+4b^2)}$.

Chapter 5: Linear Equations

- **Key Concepts**:
- Elimination / Substitution methods to find (x, y).
- Graphical representation (see Template 2 in `04_GUARANTEED_SURE_SHOT_LONG_QUESTIONS.md`).
- **Standard Solved Question**:
- *Solve for x and y: $3x+2y=11$ and $2x+3y=4$:*
  - Multiply (1) by 3: $9x+6y=33$
  - Multiply (2) by 2: $4x+6y=8$
  - Subtract: $5x=25 \implies \mathbf{x=5}$.
  - Substitute $x=5$ into (1): $3(5)+2y=11 \implies 15+2y=11 \implies 2y=-4 \implies \mathbf{y=-2}$.

Chapter 6: Quadratic Equations

- **Key Concepts**:
- $ax^2+bx+c=0$
- Discriminant: $D=b^2-4ac$
- Roots: $x=\frac{-b\pm\sqrt{D}}{2a}$
- **Standard Solved Questions**:
- *Find the discriminant and nature of roots for $2x^2-6x+3=0$:*
  - $a=2, b=-6, c=3$
  - $D=(-6)^2-4(2)(3)=36-24=\mathbf{12}$.
  - Since $D>0$, the roots are **real and distinct**.
- *Find the roots of $x^2-5x+6=0$:*
  - $(x-2)(x-3)=0 \implies \mathbf{x=2,\ 3}$.

Chapter 7: Arithmetic Progression (AP)

- **Key Concepts**:
- General term: $a_n=a+(n-1)d$
- Sum of n terms: $S_n=\frac{n}{2}[2a+(n-1)d]$ or $S_n=\frac{n}{2}[a+l]$
- **Standard Solved Questions**:
- *Which term of the AP $3, 8, 13, 18\ldots$ is 78?:*

- $a = 3, d = 5, a_n = 78$
- $78 = 3 + (n-1)5 \implies 75 = 5(n-1) \implies n - 1 = 15 \implies \mathbf{n = 16}$.

- *Find the sum of the first 20 terms of the AP* $5, 9, 13, 17 \ldots$:
  - $a = 5, d = 4, n = 20$
  - $S_{20} = \frac{20}{2}[2(5) + (20-1) \times 4] = 10[10 + 76] = 10 \times 86 = \mathbf{860}$.

MODULE 2: COMMERCIAL MATHEMATICS (8 Marks)

Chapter 9: Installment Buying

- **Key Concepts**:
- $\text{Cash Price} = \text{Down Payment} + \text{Present Value of Installments}$
- $\text{Unpaid Balance} = \text{Cash Price} - \text{Cash Down Payment}$
- **Standard Solved Question**:
- *A table is sold for ₹1,700 cash or for ₹500 as cash down payment, followed by ₹1,400 after 2 months. Find the rate of interest charged*:
  - Cash Price $= 1700$
  - Cash Down Payment $= 500$
  - Principal for the loan (Unpaid balance) $= 1700 - 500 = ₹1,200$
  - Amount paid after 2 months $= ₹1,400$
  - Interest charged for 2 months $= 1400 - 1200 = ₹200$
  - Time $T = \frac{2}{12} = \frac{1}{6}$ year
  - Using $I = \frac{P \times R \times T}{100}$:

$$200 = \frac{1200 \times R \times \frac{1}{6}}{100} = 2R \implies R = \frac{200}{2} = \mathbf{100\%} \text{ per annum.}$$

MODULE 3: GEOMETRY (25 Marks Total)

Chapter 16 & 17: Circles, Angles, Secants & Tangents

- **Key Theorems**:
- Tangent is perpendicular to radius at point of contact ($\angle OPT = 90^\circ$).
- Tangents from an external point to a circle are equal in length ($PA = PB$).
- Angle subtended by an arc at the centre is double the angle at the circumference ($\angle AOC = 2\angle ABC$).

- Angle in a semicircle is 90°.
- Opposite angles of cyclic quadrilateral sum to 180°.
- **Standard Solved Question**:
- *In the given figure, O is the centre. PA and PB are tangents to the circle from P. If $\angle APB = 60^\circ$, find $\angle AOB$:*
  - In quadrilateral $OAPB$, $\angle OAP = 90^\circ$ and $\angle OBP = 90^\circ$.
  - $\angle AOB + \angle APB = 180^\circ \implies \angle AOB = 180^\circ - 60^\circ = \mathbf{120}^\circ$.

Chapter 18: Constructions

- **Core Problem**:
- Draw a circle of radius 3 cm or 4 cm. From point P at distance d from centre, draw tangents (refer to `04_GUARANTEED_SURE_SHOT_LONG_QUESTIONS.md` Template 3).

Chapter 19: Coordinate Geometry

- **Key Formulas**:
- Distance: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
- Midpoint: $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$
- Section: $\left(\frac{mx_2+nx_1}{m+n}, \frac{my_2+ny_1}{m+n}\right)$
- **Standard Solved Question**:
- *Find the distance between $A(2, 3)$ and $B(6, 6)$:*
  - $d = \sqrt{(6-2)^2 + (6-3)^2} = \sqrt{4^2 + 3^2} = \sqrt{16+9} = \sqrt{25} = \mathbf{5}$ units.

MODULE 4: MENSURATION (10 Marks)

Chapter 21: Surface Area and Volume

- **Key Solids**:
- Cylinder: $V = \pi r^2 h,\ \text{CSA} = 2\pi rh$
- Cone: $V = \frac{1}{3}\pi r^2 h,\ l = \sqrt{r^2 + h^2},\ \text{CSA} = \pi rl$
- Sphere: $V = \frac{4}{3}\pi r^3,\ \text{Area} = 4\pi r^2$
- **Standard Solved Question**:
- *Find the volume and total surface area of a solid cylinder of radius 7 cm and height 10 cm (Take $\pi = \frac{22}{7}$):*

- $V = \pi r^2 h = \frac{22}{7} \times 7 \times 7 \times 10 = 22 \times 70 = \mathbf{1,540}\ \text{cm}^3$.
- $\text{TSA} = 2\pi r(r+h) = 2 \times \frac{22}{7} \times 7 \times (7+10) = 44 \times 17 = \mathbf{748}\ \text{cm}^2$.

MODULE 5: TRIGONOMETRY (10 Marks)

Chapter 23: Trigonometric Ratios & Heights and Distances

- **Key Trigonometric Ratios Table**: Memorize $30^\circ, 45^\circ, 60^\circ$ for $\sin, \cos, \tan$.
- **Heights & Distances**: Master the Lighthouse/Tower template (refer to Template 1 in `04_GUARANTEED_SURE_SHOT_LONG_QUESTIONS.md`).
- **Standard Solved Question**:
- *Evaluate:* $2\tan^2 45^\circ + \cos^2 30^\circ - \sin^2 60^\circ$:
  - $\tan 45^\circ = 1 \implies \tan^2 45^\circ = 1^2 = 1$
  - $\cos 30^\circ = \frac{\sqrt{3}}{2} \implies \cos^2 30^\circ = \frac{3}{4}$
  - $\sin 60^\circ = \frac{\sqrt{3}}{2} \implies \sin^2 60^\circ = \frac{3}{4}$
  - Value $= 2(1) + \frac{3}{4} - \frac{3}{4} = \mathbf{2}$.

MODULE 6: STATISTICS & PROBABILITY (12 Marks)

Chapter 25: Measure of Central Tendency

- **Direct Mean Formula**: $\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$
- **Grouped Median Formula**: $\text{Median} = L + \left(\frac{\frac{N}{2} - cf}{f}\right) \times h$
- **Empirical Mode**: $\text{Mode} = 3\,\text{Median} - 2\,\text{Mean}$

Chapter 26: Probability

- **Core Formula**: $P(E) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$, $0 \le P(E) \le 1$.
- **Standard Solved Questions**:
- *A single die is thrown. Find the probability of getting (i) a prime number, (ii) an odd number*:
  - Sample space $= \{1,2,3,4,5,6\}$ (Total = 6).
  - (i) Prime numbers $= \{2,3,5\}$ (3 outcomes) $\implies P = \frac{3}{6} = \mathbf{\frac{1}{2}}$.
  - (ii) Odd numbers $= \{1,3,5\}$ (3 outcomes) $\implies P = \frac{3}{6} = \mathbf{\frac{1}{2}}$.

- *One card is drawn from a well-shuffled deck of 52 cards. Find the probability of getting a red queen*:
    - Red Queens $= 2$ (Queen of hearts, Queen of diamonds).
    - $P = \frac{2}{52} = \mathbf{\frac{1}{26}}$.

NIOS Secondary (Class 10) Mathematics (211)

NIOS Secondary (Class 10) Mathematics (211) — Exam Prep Study Kit

NIOS Secondary (Class 10) Mathematics (211)

High-Probability Predicted Mock Examination (85 Marks)

Time: 2½ Hours | Maximum Marks: 85

> **Instructions for the Student**:
> Attempt this paper on **Day 6 (4th November)** with a pen, notebook, and geometry box.
> Set a timer for 2.5 hours. Afterwards, check your answers against the complete step-by-step solutions at the end of this document.

SECTION A (Objective Type Questions — 42 Marks)

Q1 to Q17: Multiple Choice Questions (1 Mark Each)

1. The discriminant of the quadratic equation $2x^2 - 4x + 1 = 0$ is:
   (A) 8
   (B) -8
   (C) 16
   (D) 0

2. The 12th term of the AP $3, 7, 11, 15 \ldots$ is:
   (A) 44
   (B) 47
   (C) 48
   (D) 51

3. By selling a watch for ₹660, a shopkeeper makes a profit of 10%. The cost price of the watch was:

(A) ₹600
(B) ₹590
(C) ₹550
(D) ₹726

4. In a circle of radius $5\ \text{cm}$, a tangent PT of length $12\ \text{cm}$ is drawn from external point P. The distance OP is:
   (A) $7\ \text{cm}$
   (B) $13\ \text{cm}$
   (C) $17\ \text{cm}$
   (D) $\sqrt{119}\ \text{cm}$

5. The angle subtended by a diameter at any point on the circumference of a circle is:
   (A) 45°
   (B) 60°
   (C) 90°
   (D) 180°

6. In a cyclic quadrilateral $ABCD$, $\angle B = 85^\circ$. The measure of $\angle D$ is:
   (A) 85°
   (B) 95°
   (C) 105°
   (D) 180°

7. The distance of point $P(-6, 8)$ from the origin is:
   (A) 6
   (B) 8
   (C) 10
   (D) 14

8. The coordinates of the mid-point of line segment joining $(2, -3)$ and $(6, 5)$ are:
   (A) $(4, 1)$
   (B) $(4, 2)$
   (C) $(8, 2)$
   (D) $(3, 1)$

9. The perimeter of a circle of radius $7\ \text{cm}$ is:
   (A) $22\ \text{cm}$
   (B) $44\ \text{cm}$

(C) $154\ \text{cm}$
(D) $88\ \text{cm}$

10. The volume of a cylinder whose radius is $3\ \text{cm}$ and height is $7\ \text{cm}$ is (take $\pi = \frac{22}{7}$):
    (A) $66\ \text{cm}^3$
    (B) $198\ \text{cm}^3$
    (C) $132\ \text{cm}^3$
    (D) $44\ \text{cm}^3$

11. The value of $\tan 45^\circ + \cos 60^\circ$ is:
    (A) 1
    (B) $\frac{3}{2}$
    (C) $\frac{1}{2}$
    (D) 2

12. The value of $\sec^2 45^\circ - \tan^2 45^\circ$ is:
    (A) 0
    (B) 1
    (C) 2
    (D) -1

13. The mean of $4, 6, 8, x, 12$ is 8. The value of x is:
    (A) 8
    (B) 10
    (C) 9
    (D) 7

14. The mode of the data $2, 3, 5, 3, 7, 3, 8, 2$ is:
    (A) 2
    (B) 3
    (C) 5
    (D) 7

15. If the empirical formula connects mode, median, and mean, then Mode equals:
    (A) $3\,\text{Median} - 2\,\text{Mean}$
    (B) $2\,\text{Median} - 3\,\text{Mean}$
    (C) $3\,\text{Mean} - 2\,\text{Median}$
    (D) $\text{Median} - \text{Mean}$

16. A die is thrown once. The probability of getting a number greater than 4 is:
    (A) $\frac{1}{6}$
    (B) $\frac{1}{3}$
    (C) $\frac{1}{2}$
    (D) $\frac{2}{3}$

17. If $P(E) = 0.35$, then $P(\bar{E})$ is:
    (A) 0.65
    (B) 0.75
    (C) 0.35
    (D) 1.35

Q18 to Q27: Objective Questions (2 Marks Each — 1 Mark Per Sub-Part)

1. **Fill in the blanks**:
   (i) The two factors of $x^2 - 7x + 12$ are $(x - 3)$ and `_____`.
   (ii) The value of the polynomial $3x^2 - 5x + 7$ at $x = 1$ is `_____`.

2. **True or False**:
   (i) The roots of $x^2 + 4x + 4 = 0$ are real and equal. (`TRUE / FALSE`)
   (ii) The n-th term of an AP whose first term is a and common difference is d is given by $a + nd$. (`TRUE / FALSE`)

3. **Fill in the blanks**:
   (i) A tangent to a circle intersects it in exactly `_____` point.
   (ii) The lengths of tangents drawn from an external point to a circle are `_____`.

4. **True or False**:
   (i) The angle in a semicircle is an acute angle. (`TRUE / FALSE`)
   (ii) In a cyclic quadrilateral, the sum of opposite angles is $180°$. (`TRUE / FALSE`)

5. **Fill in the blanks**:
   (i) The distance of the point $(0, 5)$ from the x-axis is `_____` units.
   (ii) The midpoint of the line segment joining $(0, 0)$ and $(4, 6)$ is `_____`.

6. **True or False**:
   (i) $\sin^2 60° + \cos^2 60° = 1$. (`TRUE / FALSE`)
   (ii) The value of $\tan 90°$ is 0. (`TRUE / FALSE`)

7. **Fill in the blanks**:
   (i) The total surface area of a cube of edge a is `_____`.
   (ii) The curved surface area of a cylinder of radius r and height h is `_____`.

8. **True or False**:
   (i) The probability of an event can never exceed 1. (`TRUE / FALSE`)
   (ii) Two coins tossed together yield 6 possible outcomes. (`TRUE / FALSE`)

9. **Installment Buying (Fill in the blanks)**:
   A bicycle has a cash price of ₹3,000. Under an installment plan, ₹600 is paid as down payment.
   (i) The unpaid balance is `_____`.
   (ii) If the remaining balance is cleared in 4 equal monthly installments of ₹650, the total interest paid is `_____`.

10. **Match the following**:
Column I	Column II
(i) $\sin 30^\circ$	(a) $\sqrt{3}$
(ii) $\tan 60^\circ$	(b) $\frac{1}{2}$

Q28: Context / Case Study Question (5 Marks — 1 Mark Per Sub-Part)

1. **Case Study**: A mathematics exhibition model displays a circular garden with centre O and radius $10\ \text{m}$. Two watchtowers are set up at points A and B along the boundary. Two straight pathways PA and PB meet at an observation gate P outside the garden. PA and PB are tangents to the circular garden, and $\angle APB = 60^\circ$.
   (i) What is the measure of $\angle OAP$?
   (ii) What is the measure of central angle $\angle AOB$?
   (iii) If $PA = 15\ \text{m}$, what is the length of pathway PB?
   (iv) What type of triangle is ΔPAB?
   (v) What is the perimeter of quadrilateral $OAPB$ if $PA = 15\ \text{m}$ and radius $= 10\ \text{m}$?

SECTION B (Subjective Questions — 43 Marks)

Q29 to Q37: Very Short Answer Questions (2 Marks Each — 18 Marks)

1. Simplify using algebraic identities: $(2x + 5y)^2 - (2x - 5y)^2$.

2. Find the discriminant and hence write the nature of the roots of the equation $3x^2 - 5x + 2 = 0$.
3. Find the 20th term of the AP $4, 9, 14, 19 \ldots$
4. Tangents PA and PB are drawn from point P to a circle with centre O. If $OP = 10\ \text{cm}$ and radius $OA = 6\ \text{cm}$, calculate the length of tangent PA.
5. In the given figure, $ABCD$ is a cyclic quadrilateral. If $\angle A = (2x + 10)^\circ$ and $\angle C = (3x + 20)^\circ$, find the value of x and the measure of $\angle A$.
6. Find the coordinates of the point which divides the line segment joining $A(1, 3)$ and $B(4, 6)$ internally in the ratio $2 : 1$.
7. Find the area of a sector of a circle of radius $6\ \text{cm}$ if the central angle is 60°. (Take $\pi = \frac{22}{7}$).
8. Evaluate: $\frac{5\cos^2 60^\circ + 4\sec^2 30^\circ - \tan^2 45^\circ}{\sin^2 30^\circ + \cos^2 30^\circ}$.
9. A card is drawn at random from a pack of 52 playing cards. Find the probability that the card drawn is:
   (i) A king of red colour
   (ii) A spade

Q38 to Q42: Short Answer Questions (3 Marks Each — 15 Marks)

1. Factorise: $x^3 - 8y^3$.
2. Find the ratio in which the point $P(-1, 6)$ divides the line segment joining the points $A(-3, 10)$ and $B(6, -8)$.
3. Construct a circle of radius $3.5\ \text{cm}$. From a point P at a distance of $6.5\ \text{cm}$ from its centre, construct a pair of tangents to the circle.
4. A solid metallic cylinder of radius $4.2\ \text{cm}$ and height $10\ \text{cm}$ is melted and recast into a sphere. Find the radius of the sphere.
5. Calculate the median of the following frequency distribution:

Class Interval	$0 - 10$	$10 - 20$	$20 - 30$	$30 - 40$	$40 - 50$
Frequency (f)	5	8	15	12	10

Q43 to Q44: Long Answer Questions (5 Marks Each — 10 Marks)

1. Solve the following system of linear equations graphically:

$$x + y = 5$$

$$2x - y = 4$$

Find the vertices of the triangle formed by these two lines and the y-axis.

2. A vertical pole of height $60\ \text{m}$ stands on horizontal ground. From the top of the pole, the angles of depression of two cars parked on opposite sides of the pole are observed to be 30° and 45°. Find the distance between the two cars. (Take $\sqrt{3} \approx 1.732$).

COMPLETE STEP-BY-STEP SOLUTIONS & MARKING SCHEME

Section A Solutions

- **Q1**: (A) 8 [$D = (-4)^2 - 4(2)(1) = 16 - 8 = 8$]
- **Q2**: (B) 47 [$a_{12} = 3 + (11)(4) = 47$]
- **Q3**: (A) ₹600 [$CP = \frac{660}{1.1} = 600$]
- **Q4**: (B) $13\ \text{cm}$ [$OP = \sqrt{5^2 + 12^2} = 13$]
- **Q5**: (C) 90°
- **Q6**: (B) 95° [$180 - 85 = 95$]
- **Q7**: (C) 10 [$\sqrt{(-6)^2 + 8^2} = 10$]
- **Q8**: (A) $(4, 1)$ [$(\frac{2+6}{2}, \frac{-3+5}{2}) = (4, 1)$]
- **Q9**: (B) $44\ \text{cm}$ [$2 \times \frac{22}{7} \times 7 = 44$]
- **Q10**: (B) $198\ \text{cm}^3$ [$\frac{22}{7} \times 9 \times 7 = 198$]
- **Q11**: (B) $\frac{3}{2}$ [$1 + \frac{1}{2} = \frac{3}{2}$]
- **Q12**: (B) 1 [Identity $\sec^2\theta - \tan^2\theta = 1$]
- **Q13**: (B) 10 [$\frac{4+6+8+x+12}{5} = 8 \implies 30 + x = 40 \implies x = 10$]
- **Q14**: (B) 3 [Most frequent]
- **Q15**: (A) $3\,\text{Median} - 2\,\text{Mean}$
- **Q16**: (B) $\frac{1}{3}$ [Numbers are $5, 6 \implies \frac{2}{6} = \frac{1}{3}$]
- **Q17**: (A) 0.65 [$1 - 0.35 = 0.65$]
- **Q18**: (i) $(x - 4)$ | (ii) 5 [$3(1) - 5(1) + 7 = 5$]
- **Q19**: (i) TRUE [$D = 16 - 16 = 0$] | (ii) FALSE [Formula is $a + (n-1)d$]
- **Q20**: (i) one | (ii) equal
- **Q21**: (i) FALSE [It is 90°] | (ii) TRUE
- **Q22**: (i) 5 | (ii) $(2, 3)$
- **Q23**: (i) TRUE | (ii) FALSE [$\tan 90^\circ$ is not defined]

- **Q24**: (i) $6a^2$ | (ii) $2\pi rh$
- **Q25**: (i) TRUE | (ii) FALSE [4 outcomes: HH, HT, TH, TT]
- **Q26**: (i) ₹2,400 [$3000 - 600$] | (ii) ₹200 [$4 \times 650 = 2600 \implies 2600 - 2400 = 200$]
- **Q27**: (i) $\rightarrow$ (b) $\frac{1}{2}$ | (ii) $\rightarrow$ (a) $\sqrt{3}$
- **Q28**: (i) 90° | (ii) 120° [$180 - 60$] | (iii) $15\ \text{m}$ [$PA = PB$] | (iv) Equilateral triangle [Since $PA = PB$ and $\angle P = 60^\circ$] | (v) $50\ \text{m}$ [$15 + 15 + 10 + 10 = 50$]

Section B Detailed Solutions

Q29:
Use identity $(a+b)^2 - (a-b)^2 = 4ab$.
Here $a = 2x, b = 5y \implies 4(2x)(5y) = \mathbf{40xy}$.

Q30:
$a = 3, b = -5, c = 2$.
$D = (-5)^2 - 4(3)(2) = 25 - 24 = \mathbf{1}$.
Since $D > 0$, the roots are **real and distinct (unequal)**.

Q31:
$a = 4, d = 5, n = 20$.
$a_{20} = a + (20-1)d = 4 + 19(5) = 4 + 95 = \mathbf{99}$.

Q32:
In right ΔOAP, $\angle OAP = 90^\circ$.
By Pythagoras: $PA = \sqrt{OP^2 - OA^2} = \sqrt{10^2 - 6^2} = \sqrt{100 - 36} = \sqrt{64} = \mathbf{8}\ \text{cm}$.

Q33:
Opposite angles sum to 180°:
$(2x + 10) + (3x + 20) = 180 \implies 5x + 30 = 180 \implies 5x = 150 \implies \mathbf{x = 30}$.
$\angle A = 2(30) + 10 = \mathbf{70}^\circ$.

Q34:
$m : n = 2 : 1$, $A(1,3), B(4,6)$.
$x = \frac{2(4)+1(1)}{2+1} = \frac{8+1}{3} = \frac{9}{3} = 3$.
$y = \frac{2(6)+1(3)}{2+1} = \frac{12+3}{3} = \frac{15}{3} = 5$.
Point is $\mathbf{(3, 5)}$.

Q35:
$\text{Area} = \frac{\theta}{360}\pi r^2 = \frac{60}{360} \times \frac{22}{7} \times 6 \times 6 = \frac{1}{6} \times \frac{22}{7} \times 36 = \frac{132}{7} \approx \mathbf{18.86}\ \text{cm}^\mathbf{2}$.

Q36:
Denominator $= \sin^2 30^\circ + \cos^2 30^\circ = 1$.
Numerator $= 5\left(\frac{1}{2}\right)^2 + 4\left(\frac{2}{\sqrt{3}}\right)^2 - (1)^2 = 5\left(\frac{1}{4}\right) + 4\left(\frac{4}{3}\right) - 1 = \frac{5}{4} + \frac{16}{3} - 1 = \frac{15+64-12}{12} = \mathbf{\frac{67}{12}}$.

Q37:
(i) Red Kings $= 2 \implies P = \frac{2}{52} = \mathbf{\frac{1}{26}}$.
(ii) Spades $= 13 \implies P = \frac{13}{52} = \mathbf{\frac{1}{4}}$.

Q38:
$x^3 - (2y)^3 = (x - 2y)(x^2 + 2xy + (2y)^2) = \mathbf{(x - 2y)(x^2 + 2xy + 4y^2)}$.

Q39:
Let ratio be $k : 1$.
Using x-coordinate: $-1 = \frac{k(6)+1(-3)}{k+1} \implies -(k+1) = 6k - 3 \implies -k - 1 = 6k - 3 \implies 7k = 2 \implies k = \frac{2}{7}$.
Ratio is $\mathbf{2 : 7}$.

Q40:
Radius $r = 3.5$ cm, $OP = 6.5$ cm.
Length of tangent $= \sqrt{6.5^2 - 3.5^2} = \sqrt{(6.5 - 3.5)(6.5 + 3.5)} = \sqrt{3 \times 10} = \sqrt{30} \approx \mathbf{5.48}$ cm.

Q41:
Volume of cylinder $=$ Volume of sphere
$\pi r_1^2 h = \frac{4}{3}\pi r_2^3 \implies (4.2)^2 \times 10 = \frac{4}{3} r_2^3$
$17.64 \times 10 = \frac{4}{3} r_2^3 \implies 176.4 \times \frac{3}{4} = r_2^3 \implies 132.3 = r_2^3 \implies r_2 \approx \mathbf{5.1}$ cm.

Q42:
Total $N = 5 + 8 + 15 + 12 + 10 = 50 \implies \frac{N}{2} = 25$.
Cumulative frequencies: $5, 13, 28, 40, 50$.
Median class is $20 - 30$ ($L = 20, cf = 13, f = 15, h = 10$).
$\text{Median} = 20 + \frac{25-13}{15} \times 10 = 20 + \frac{12}{15} \times 10 = 20 + 8 = \mathbf{28}$.

Q43:
- Equation 1 ($x + y = 5$): $(0, 5), (5, 0), (2, 3)$.
- Equation 2 ($2x - y = 4$): $(0, -4), (2, 0), (3, 2)$.
Intersection point: $x = 3, y = 2 \implies \mathbf{(3, 2)}$.
Vertices of triangle formed with y-axis ($x = 0$):
Line 1 hits y-axis at $(0, 5)$; Line 2 hits y-axis at $(0, -4)$; Intersection is $(3, 2)$.
Vertices: $\mathbf{(0, 5)}$, $\mathbf{(0, -4)}$, $\mathbf{(3, 2)}$.

Q44:

Let pole be $AB = 60\ \text{m}$.

- Car 1 at C (elevation 45°): $\tan 45^\circ = \frac{60}{BC} \implies BC = 60\ \text{m}$.
- Car 2 at D (elevation 30°): $\tan 30^\circ = \frac{60}{BD} \implies \frac{1}{\sqrt{3}} = \frac{60}{BD} \implies BD = 60\sqrt{3} = 60 \times 1.732 = 103.92\ \text{m}$.

Total distance between cars:

$CD = BC + BD = 60 + 103.92 = \mathbf{163.92}\ \text{m}$.